

OPJ Girls College F-8/2, Islamabad

First Term EXAMINATION 2026

SUBJECT: Physics

CLASS: MII



Invigilator Sign:

Candidate Name Ikram Abif

Class: M-II Section: H

Q.1. Fill the relevant bubble against each question according to curriculum:

Question	سوال	A	B	C	D	A	B	C	D
1. How many significant zeroes are in 0.00305?		1	2	3	4	☐	☐	☒	☐
2. Which is a systematic error?		A misread scale	Random disturbances in the lab	An unpredictable fluctuation in readings	Different readings due to human reaction time	☐	☐	☒	☐
3. A rider is training a horse that moves 60m towards right in 3 sec. Then it returns 30m in 2 sec. His average velocity is		6 m/s	18 m/s	35 m/s	Zero	☒	☐	☐	☐
4. A ball is thrown vertically upwards. Its velocity at highest point is		10 m/s	Zero	10 m/s ²	-10m/s ²	☐	☐	☒	☐
5. The unit of momentum is		Nm	Kgm/s ²	Ns	N/s	☐	☐	☒	☐
6. When horse pulls a cart, the action is on the,		Cart	Earth	Horse	Earth and cart	☒	☐	☐	☐
7. Which of the following is an example of non contact force?		Friction	Tension	Magnetic force	Normal force	☐	☐	☒	☐
8. The centre of gravity of a triangular shaped body is		Point of intersection of its medians	Point of intersection of altitudes	The centre of triangle	Point of intersection of its diagonals	☐	☐	☒	☐

9.If mass of a body is halved and velocity is doubled, then its K.E will be	$\frac{1}{2} mv^2$	mv^2	$2mv^2$	$\frac{1}{4}mv^2$	☐ ☐ ☒ ☐
10.Work done will be zero if the angle between force and displacement is	0°	45°	90°	180°	☐ ☐ ☒ ☐
11.The energy stored in a dam is	Electrical energy	Potential energy	Kinetic energy	Thermal energy	☒ ☐ ☐ ☐

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Signature Of Invigilator

[Signature]

Copy No:

Name: Chenab Atif

Roll No. 24683

Class: M.2

Section: 11

Subject: Physics

Date: 3.9.2026

No Of Continuation Sheets Attached

Question No.	1	2	3	4	5	6	7	8	9	10	11	12	Total	Out of
Marks Obtained		4	2	6	6								27	50

Start writing here

Section - B

Q.2 (i)

A.2 (i)

① 1168×10^{-27}

$1.168 \times 10^3 \times 10^{-27}$

1.168×10^{-24}

② 725×10^5

$7.25 \times 10^2 \times 10^5$

7.25×10^7

③ 0.02×10^{-8}

0.02×10^{-8}

$2.0 \times 10^{-2} \times 10^{-8}$

2.0×10^{-10}

Q.2 (ii) (or)

A.2 (ii) (or)

given:

mass = 600N {weighs}

$g = 1.62 \text{ m/s}^2$

$w = ?$

formula: $F = W = mg$

$$W = 600 \times 9.8162$$
$$= 972 \text{ N}$$

Q.2 (iii)

A.2 (iii)

Distance

Displacement

Defination

- * The path travelled between 2 points initial point to final point is called distance
- * The shortest distance travelled between 2 points is called displacement

Symbolism

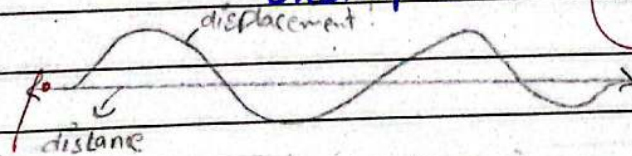
* It's symbol is s .

* It's symbol is $\vec{s}$.

Quantity

- * It belongs to a scalar quantity since there is no direction.
- * It belongs to vector quantity since there is a direction.

Example



Q.2 (iv) (or)

A.2 (iv) (or)

wearing helmet is safe while driving motorcycle ~~between~~ because if we are in severe accident helmet will protect our head. and it will not

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Name Adnan Afif

Class / Section M-2 H

Examination First term exams.



Roll # 24683

Subject Physics

Date _____

Q. #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	TOTAL
Marks Obtained															

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and ~~if~~ if we ~~were~~ just crashed with other truck and we will fall from the motorcycle. if we didn't wear any helmet then it will a lot of damage. but if we wore the helmet then it will save us from severe damage.

①

Reason.

Q:2(v)(or)

A:2(v) (or)

Rolling friction

Sliding friction.

Defination

* Rolling friction is ~~produce~~ ^{is produced} friction that ~~rolls~~ ^{is produced} on the ground and friction is made.

Sliding friction is a friction that slides on the ground and friction is made.

②

for example:- Rolling a football

for example:- Sliding doors the friction is made b/w door and ground.

Q:2(vi)(or)

A:2(vi) (or)

[P.T.O.]

given :-

$$d_1 = 20\text{cm}$$

converting cm into m :-

$$\frac{20\text{cm}}{100}$$

$$= 0.2\text{m}$$

$$F_1 = 50\text{N}$$

$$F_2 = 100\text{N}$$

$$d_2 = ?$$

formula torque :- $d_1 \times F_1 = d_2 \times F_2$

$$0.2 \times 50 = d_2 \times 100$$

$$\frac{0.2 \times 50}{100} = d_2$$

$$0.1\text{m} = d_2$$

Q:2(vii)(or)

A:2(vii)(or)

170 km/hr into m/s

$$170 \times \frac{1000}{3600}$$

$$= 47.2\text{ m/s}$$

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Name: aseemab Afif

Roll No. 24683

Class: M-2

Section: H

Subject: Physics

Date: 2-9-2026

No Of Continuation Sheets Attached

Question No.	1	2	3	4	5	6	7	8	9	10	11	12	Total	Out of
Marks Obtained														

Start writing here

(b) 1 m/s^2 into km/hr^2

$$1 \times \frac{(3600)^2}{1000}$$

$$= 12960 \text{ km/hr}^2$$

(c) 36 m/s into km/hr

$$36 \times \frac{3600}{1000}$$

$$= 129.6 \text{ km/hr}$$

Q:2(vii)

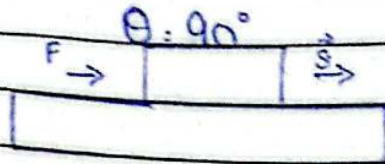
A:2(vii)

* Maximum workdone:- when force and displacement is parallel to each other.

$$\begin{aligned} & \theta = 0^\circ \\ & \left. \begin{aligned} & F s (\cos^\circ) (\theta) \\ & F s (\cos^\circ) (0) \end{aligned} \right\} \begin{aligned} & F s (\cos) (1) \\ & = F s \end{aligned} \end{aligned}$$

* Perpendicular workdone -

when force and displacement are perpendicular to each other.



$$FS(\cos)(90)$$

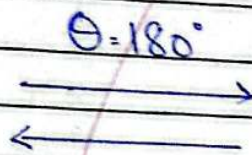
$$FS(0)$$

$$0$$

Q.3

* Minimum workdone -

when force and displacement are not parallel to each other.



$$= FS(\cos)(180)$$

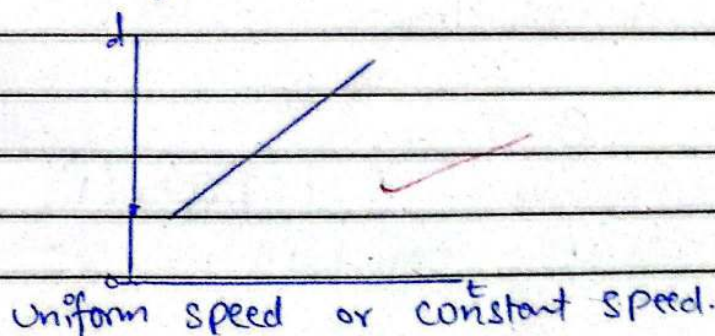
$$= FS(-1)$$

$$= -FS$$

$$Q: 2(\text{in})(\text{or})$$

$$A: 2(\text{in})(\text{or})$$

@ body moving with constant speed.



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Copy No:

Name: Cheema Abif

Roll No. 24683

Class: M-2

Section: H

Subject: Physics

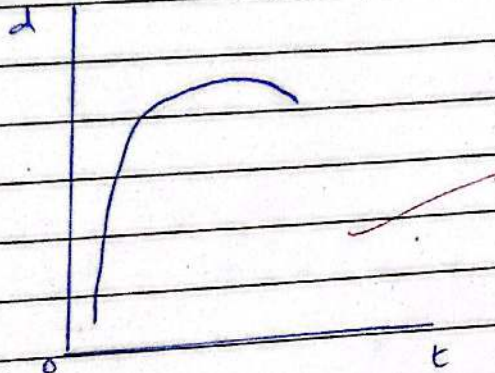
Date: 3-9-2026

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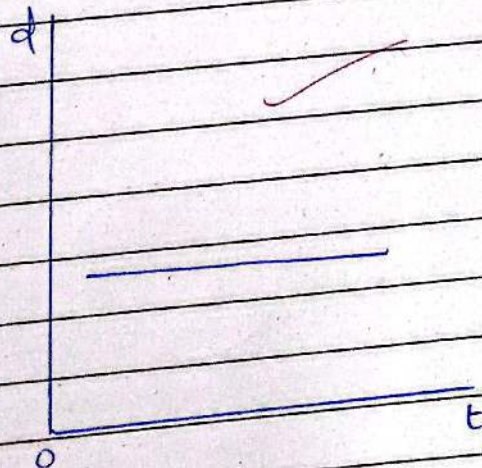
② body moving with decreasing speed.



deceleration

③ body at rest

③



Section - C

Q:3 (or)

A:3 (or)

given :-

distance :- 800 m

Δ time taken :- 40 s

displacement : 600 m

Δ time taken :- 36 s

① Average speed :-

$$\text{Speed} = v = \frac{\text{distance}}{\text{time}}$$

$$= \frac{800}{40}$$

$$= 20 \text{ m/s}$$

② Average velocity :-

$$\text{velocity} = \vec{v} = \frac{\text{displacement}}{\text{time}}$$

$$= \frac{600}{36}$$

$$= 16.67 \text{ m/s}$$

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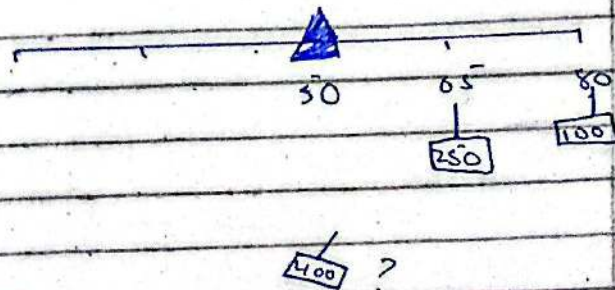
Examiner _____

Rechecked by _____

Q: 4

A: 4

Diagram rough { for understanding }



given :-

distance $\therefore \frac{65 - 50}{1} = 15 \text{ cm}$

mass₁ = 250g

$d_1 = 80 - 50 = 30 \text{ cm}$

mass₂ = 100g

formula $\therefore m_3 \times d_3 = (m_1 d_1) + (m_2 d_2)$

let d_3 be x

$$400g(x) = \cancel{15 \times 250} (250 \times 15) + (100 \times 30)$$

$$400x = 3750 + 3000$$

$$400x = 6750$$

$$x = \frac{6750}{400}$$

$$x = 16.875 \text{ cm}$$